MILITARY SPECIFICATION SHEET

STARTER, AIRCRAFT ENGINE, HYDRAULIC FOR T58 ENGINE

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the starter described herein shall consist of this document and the latest issue of Specification MIL-S-22999.

REQUIREMENTS

Envelope: Figures 1 and 2; must fit diametrically within Figure 3.

Wt. max dry (Lb): 8.0

Aircraft: Model CH-46

Engine: Model T58

Engine accessory drive: The starter shall be suitable for mounting on an AND20009, Type XIX drive modified to conform with the dimensions shown in MIL-S-7780. For optional output spline drive starters, a female spline is provided in the engine accessory drive. Where,conflicts occur, the values specified herein will be the governing specification.

The mating male spline on the starter output drive shaft shall be in accordance with the following Society of Automotive Engineers (SAE) standard spline data:

Number of teeth	16
Diametral Pitch	20/40
Pitch Diameter	0.800 inches
Pressure Angle	30 degrees
TIF Diameter	0.750 inches (maxinim)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: ENGINEERING SPECIFICATIONS AND STANDARDS DEPARTMENT (CODE 93), NAVAL AIR ENGINEERING CENTER, LAKEHURST, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 2995

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Minor Diameter	0.726/0.736	inches
Major Diameter	0.845/0.850	inches
Tooth Thickness Circular	0.072/0.077	inches
Circular Pitch	0.1571	

The male starter spline shall extend .971 inches maximum from the starter mounting flange face and shall have .430 ± .010 inches for the overall length of the spline teeth. A clearance hole shall be provided in the end of the starter spline with a minimum diameter of .540 inches and a minimum depth from the spline end of .187 inches.

Rotation viewed from anti-output shaft end: Counterclockwise

Ports: Conform to: Inlet - MS33649-8; Outlet - MS33649-8

Ambient temperature operating range (oF): -65 to +275

Exposure temperature range (°F): -103 to +275

Overspeed limit, min peak no-load speed (rpm): 13,000

Inlet pressure, max at zero flow (psig): 4400

Rated inlet pressure (psig): 3750; +50, -o

Rated outlet pressure (psig): 50; +15, -o

Rated inlet fluid temperature (°F): +275; +0, -10

Rated fluid flow (gpm): 9.0; maximum allowable flow at no-load operation is 9.0 gpm.

Rated torque (lb-ft): 12 at 3700 psig differential pressure, 8.5 at 2750 differential pressure.

Rated drive shaft speed (rpm): 6000

Automatic shutoff control: Cutout switch shall be normally open; it shall close upon application of hydraulic oil pressure and open at cutout speed. The switch shall withstand a surge current of 2 amperes at 28 volts inductive load at -65°'F.

Drive shaft cutout speed (rpm): 12,000; +500, -500

Initial calibration:

Hydraulic fluid flow, max (gpm):
Zero to max cutout drive speed: 9.0
No-load drive speed: 9.0

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Performance:

Minimum Starter Drive Torque (Lb-Ft)
and Corresponding Starter Drive Speeds (RPM)

Pressure Differential (PSIG)	Speed	Torque	Speed	Torque	Speed	Torque	Speed	Torque
3,700	0	10	1,000	12	6,000	12	11,600	3
2,750	0	7	1,000	8.5	6,000	8.5	11,600	2

No-load operation: The starter shall be operated for two minutes. The starter free-running-speed obtained during this test, or the maximum speed attained before automatic cutback, shall not be less than 13,000 rpm.

Proof pressure test: With the loading device adjusted to limit starter output drive speed ,to rated speed and at rated inlet temperature, the unit shall be operated for five minutes at 5000-psig inlet pressure and at 500-psig outlet pressure. There shall be no leakage from the starter sufficient to form a drop, except from the output shaft seal from which a maximum leakage of 5 cc per hour is allowable.

Endurance test: The starter shall be subjected to a 1200-cycle endurance test as outlined herein. The following schedule for fluid inlet temperature and fluid inlet pressure shall be maintained:

Number of Cycles	Inlet Fluid Temperature 'F	Inlet Fluid Pressure PSIG
400	275	3750
600	220	3900
200	160	3750

All tests shall be conducted at rated outlet pressure. Each cycle shall consist of three consecutive phases as follows:

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Test Stand Rotor Polar Moment of Inertia	Phase A - Acceleration		Phase B - 10 sec at Following Conditions		Phase C - Acceleration	
	Drive Speed RPM	Time	Drive Speed RPM	Load Torque	Drive Speed RPM	Time
Lb-Ft2	(Min)	Sec	(Min)	Lb-Ft	(Min)	S ec
2.6	6000	5	6000	12 +1-0	12,000	9

NOTE : The above values do not allow for flywheel windage losses, test stand drag, or inertia of the starter. In order to determine compliance with this specification, the test equipment must be calibrated and specific test results corrected accordingly. If the acceleration specified for Phase C is not developed because of test stand drag torque, the test stand inertia load may be reduced or external power may be applied to assist acceleration to cutout speed. Time for end of Phase C to starter cutout shall not exceed 15 seconds.

Sustained motoring: Operational time, six minutes.

Overrunning test: The starter output shaft shall be driven for 1000 hours in a counterclockwise direction (viewed from the anti-output shaft end) in accordance with the following tabulation:

Starter Drive Overrunning Speed - RPM

Phase A 750 Hours	Phase B 200 Hours	Phase C 50 Hours
25,600	26,300	15,800

After a maximum of each 25 hours of continuous overrunning, . output shaft rotation shall be stopped for at least 15 minutes.

Low temperature operation: During Phase B of this test a minimum torque of 8.2 lb-ft at a drive speed of 6000 rpm shall be developed.

Dynamic torque check: The maximum peak transient torque shall be 28 lb-ft.

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Shear section strength, max static torque (lb-ft): 28

Shaft seal leakage (cc per hour): 1 at 26,300 rpm overrunning speed.

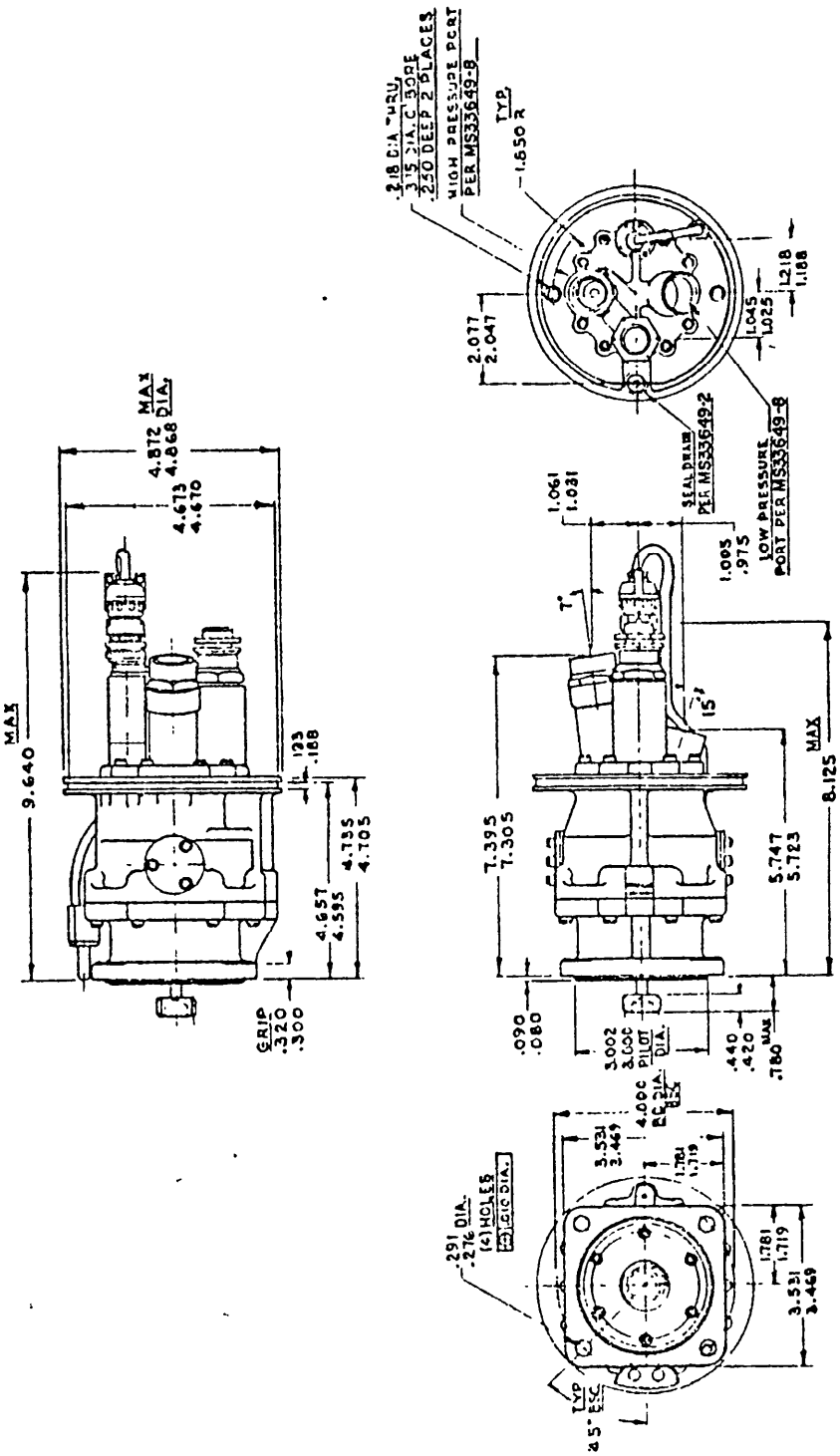
Over pressurization pressure (psig): 4500

Part number: The part number shall consist of the basic number of this specification sheet.

Example: M22999/1

Preparing Activity
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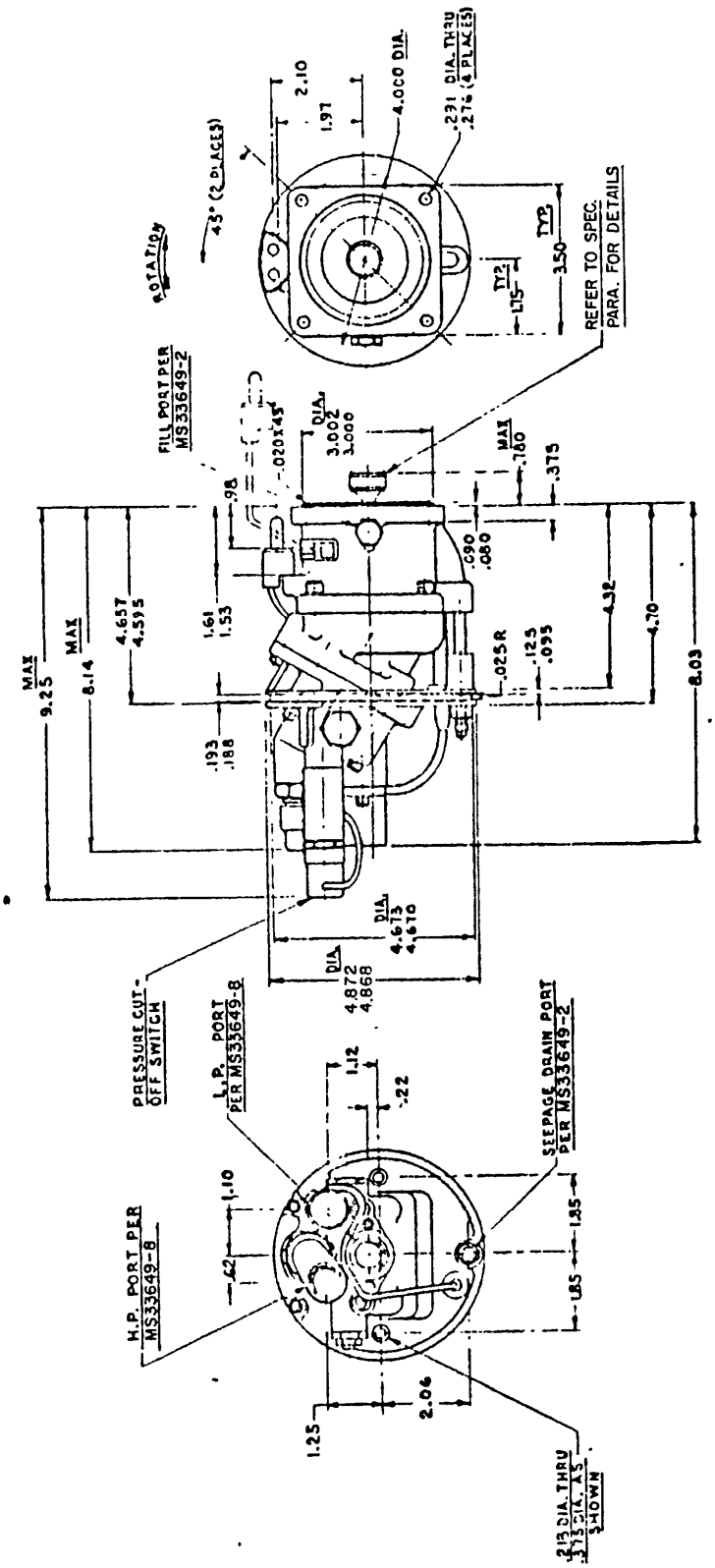
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NOTES:
1. DIMENSIONS IN INCHES UNLESS OTHERWISE SPECIFIED

FIGURE 1. STARTER ENVELOPE

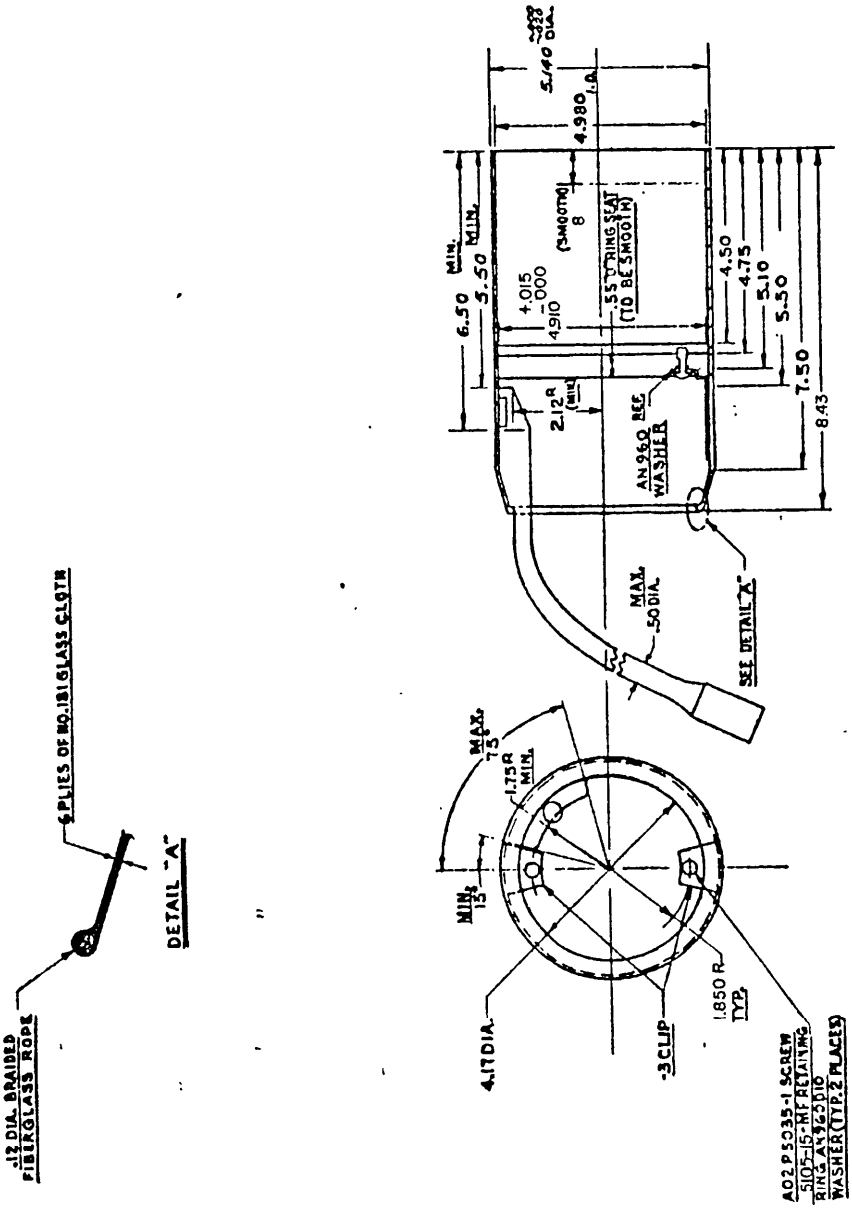
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FIGURE 2. STARTER ENVELOPE (ALTERNATE CONFIGURATION)

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FIGURE 3 COVER ASSEMBLY

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STARTER, AIRCRAFT ENGINE, HYDRAULIC FOR T58 ENGINE

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